

ASTYAN MARTIN

Portfolio link : <https://astyanm.github.io/portfolio/fr>

PROFESSIONAL EXPERIENCE

March 2026 - July 2026

MBDA France,
Le Plessis-Robinson

AI & Embedded Software Developer (RAG)

Designed an intelligent assistance solution (RAG) to optimize low-level embedded software development.

- **RAG Architecture & Data Ingestion:** Designed a hierarchical indexing pipeline for documentation and source code (C, ARM ecosystem).
- **AI Evaluation & Metrics:** Utilized the Ragas framework to evaluate model performance on critical hardware issues.

Sept. 2025 - Feb. 2026

Marianne Education,
Boulogne-Billancourt

AI & Automation Developer

Designed and deployed intelligent applications and guidance tools for the education sector, especially the students.

- **Full-Stack & Scalability:** Architected and deployed a university admission simulator (Parcoursup), maintaining a 50k+ lines codebase.
- **Impact & Traffic:** Supported 200k+ simulations and 60k+ choices created, generating 10k+ leads (50% via a Top 1 organic SEO ranking).
- **NLP & Fine-Tuning:** Trained and fine-tuned a CamemBERT model achieving 93% accuracy on a custom dataset of 320 documents.

EDUCATION

Feb 2025 - June 2025

Trento, Italy

Università di Trento

Erasmus semester focused on Artificial Intelligence and Data Science.

2023 - 2027

Lille, France

École Centrale de Lille

Master's Degree in Engineering, specializing in Data Science and Artificial Intelligence.

2021 - 2023

Strasbourg, France

Lycée Kléber (CPGE PC*)

Intensive theoretical preparation for competitive entrance exams of engineering schools.

2018 - 2021

Colmar, France

Lycée Bartholdi

High School Diploma (Baccalauréat), Highest Honors, majoring in Mathematics & Physics.

TECHNICAL PROJECTS

Medical Data NLP Extraction (EHR) | CRISTAL Lab & GIP SIB

- **LLM Pipeline:** Worked within a 12-person team to structure 1,000+ patient records.
- **Machine Learning:** Developed an XGBoost model for predictive criticality triage and length of stay prediction using PCA/StandardScaler.

Open Whisper: Real-Time Speech-to-Text Application | *Personnal Project*

- **Developed** a 100% local desktop application (Tauri + Whisper), achieving ~1s latency on GPU and 40x faster-than-real-time processing on audio files.
- **Reliability:** Secured the architecture with 115 automated tests (96 backend, 19 frontend).

SKILLS

- **Artificial Intelligence & Data:** Machine Learning, Deep Learning (PyTorch), NLP (LLM, RAG, Hugging Face), Computer Vision, Data Science (Pandas, Scikit-Learn, XGBoost, SQL).
- **Software & Web Development:** Python, TypeScript, React, Vue.js, FastAPI, Full-Stack Architecture, C.
- **Tools & Engineering:** Git, Docker, CI/CD (GitHub Actions), Embedded Systems (Raspberry Pi, Signal Processing).

INTERESTS

- **Sports:** Trail running, rock climbing, and volleyball. High-level competitive wrestling.
- **Photography:** Participated in several competitions.
- **Driver's License** (Category B)

LANGUAGES

- **English:** Full professional proficiency (C1, TOEIC: 975/990).
- **German & Chinese:** Beginner proficiency.